

## ELECTIVE COURSE-7.1

| URA414: ADDITIVE MANUFACTURING                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |   |   |     |
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| <p><b>Course Objective:</b> This course introduces the basic fundamentals of additive manufacturing (AM), its fabrication methodology, different techniques of part fabrication, materials, and various areas of defects and improvements in AM. The course also introduces the concept of reverse engineering.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |   |   |     |
| <p><b>Syllabus</b></p> <p><b>Introduction:</b> Classification of manufacturing processes, Introduction to additive manufacturing (AM), Principles of AM, Steps in AM, Advantages of AM, Classifications of Different AM Techniques: Based on raw material, Based on layering technique and energy sources.</p> <p><b>Design of CAD Models for AM:</b> Transformations, Design of curves, Solid modeling for AM.</p> <p><b>STL files for AM:</b> STL file generation, Defects in STL files and their repairs, other Interface formats, Tessellation: Voronoi Diagram, Incremental Algorithm of Triangulation, and Watson's algorithm for Delaunay triangulation.</p> <p><b>Process Technology in AM:</b> Stereo-lithography (SL), Selective Laser Sintering, Selective Laser Melting, Ballistic Particle Manufacturing, Fused Deposition Modeling, Laminated Object Manufacturing, Paper Lamination Technology, Solid Ground Curing, Beam Interference Solidification, Binder Jetting, Laser Engineered Net Shaping (LENS), 3D Printing.</p> <p><b>Problem Areas in AM:</b> Study of Slicing methods &amp; design of support structures, Part deposition orientation studies, the study of volumetric shrinkage and dimensional accuracy.</p> <p><b>Reverse Engineering:</b> Introduction to Reverse Engineering (RE), Measuring Devices, CAD Model Construction from Point Clouds, Data Handling and Reduction Methods, Applications of RE.</p> |   |   |   |     |
| <p><b>Laboratory Work (if applicable)</b></p> <ol style="list-style-type: none"> <li>1. To generate Solid Models with the given dimensions using s/w like Pro-E or Solid Works.</li> <li>2. To fabricate a prototype in AM Facility after removing STL file defects.</li> <li>3. To develop dimensional inaccuracy compensation factors for PLA/ABS and liquid resin developed prototypes.</li> <li>4. To study the effects of part deposition orientation on the dimensional inaccuracy of PLA/ABS and liquid resin developed parts.</li> <li>5. Study the effects of curling on volumetric shrinkage on PLA and ABS-developed prototypes.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |   |   |     |

*Approved in 109<sup>th</sup> meeting of the Senate held on March 16, 2023. Revised in 112<sup>th</sup> and 113<sup>th</sup> meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1<sup>st</sup> meeting of Academic Council held on September 11, 2025.*

6. To measure and evaluate the effect of process parameters on surface roughness and edge profile of PLA/ABS and liquid resin printed parts.
7. The students will be doing a project (a fully functional assembly) realizing the application of AM technology for product development.

### **Course Learning Outcomes (CLOs)**

The students will be able to:

1. Implement the concepts of transformation and solid modeling in developing a solid model.
2. Examine & detect the errors in .stl files and implement a suitable repair algorithm.
3. Identify suitable process and process parameters required for fabricating a part through additive manufacturing.
4. Evaluate the effect of slicing methods, design of support structures, and part deposition orientation on volumetric shrinkage and accuracy of the developed parts.

### **Text Books**

1. Chua, C.K., Leong, K.F., Rapid Prototyping: Principles and Applications in Manufacturing, John Wiley and Sons Inc., 2000.
2. Pham, D.T., Demov, S.S., Rapid Manufacturing: The Technologies and Applications of Rapid Prototyping and Rapid Tooling, Springer-Verlag London Limited, 2001.
3. Saxena, A., Sahay, B., Computer Aided Engineering Design, Anamaya Publishers, New Dehi, 2005.
4. Noorani, R., 3D Printing Technology, Applications, and Selection, CRC Press, Taylor & Francis Group, 2018.

### **Reference Books**

1. Patri, K. V., Weiyin, Ma, Rapid Prototyping - Laser-based and Other Technologies, Kluwer Academic Publishers, U.S.A., 2003.
2. Noorani, R., Rapid Prototyping: Principles and Applications, John Wiley & Sons, Inc., New Jersey, 2006.
3. Hopkinson, N., Hague, R.J.M., Dickens, P.M., Rapid Manufacturing- An Industrial Revolution for the Digital Age, John Wiley & Sons Ltd., U.K., 2006.
4. Zeid, I., Mastering CAD/CAM, Tata McCraw Hill, 2006.

### **Evaluation Scheme:**

| S. No. | Evaluation Elements  | Weightage |
|--------|----------------------|-----------|
| 1      | MST                  | 35        |
| 2      | EST                  | 25        |
| 3      | Sessional (Quizzes)  | 10        |
| 4      | Lab Work and Project | 30        |

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